

Set 15

Café Royale	
Item	Price (€)
Tea	1.80
Coffee	2.40
Gateau	2.95
Pastry	1.90

Q1. You make 80% profit on each coffee you sell but only 60% on pastry. When you sell coffee and pastry together you give a discount of 10%. How much profit do you make on each coffee and pastry sale?

- A. €2.63
- B. €2.88
- C. €3.01
- D. €3.44
- E. €3.87

Q2. The café sells 10 slices of gateau which have a profit margin of 40% and it sells twice as many pastries which have a profit margin 40% higher than the gateau. How much profit do you make on these items?

- A. €17.88
- B. €21.28
- C. €30.40
- D. €33.08
- E. €42.20

Q3. Your costs rise and coffee, which had a profit margin of 70%, is now down to 62%. By what percentage have your costs on coffee risen?

- A. 8%
- B. 11.4%
- C. 12.9%
- D. 21.1%
- E. 26.7%

Set 16

Marathon Results 2010 (42.2 km)	
Runner	Time (h:mm:ss)
Anderson	2:12:14
Benoit	2:12:38
Canard	2:13:31
Dennis	2:14:12
Ekimov	2:14:20

Q1. Canard passed the halfway marker at 1:07:50. What was his average speed for the second half of the race?

- A. 15.20 km/h
- B. 17.38 km/h
- C. 18.63 km/h
- D. 19.27 km/h
- E. 20.31 km/h

Q2. Dennis and Ekimov passed the "1km to go" marker at exactly the same time, 2:11:00. How much faster than Ekimov did Dennis run in the last kilometre?

- A. 1%
- B. 4.2%
- C. 6.6%
- D. 8%
- E. 12.5%

Q3. Benoit ran the last 150m at a speed 5% faster than his overall average for the race. How far was Benoit from the finishing line when Anderson finished?

- A. 84m
- B. 92m
- C. 100m
- D. 133m
- E. 149m

Set 17

Acme Closing Share Prices (€)		
	Buy	Sell
Monday	2.37	2.32
Tuesday	2.41	2.37
Wednesday	2.39	2.34
Thursday	2.28	2.25
Friday	2.34	2.29

Q1. You buy 2000 shares on Tuesday at 1.4% below the previous day's closing price. There is a €15.95 dealing charge. How much does it cost you?

- A. €4673.64
- B. €4689.59
- C. €4734.37
- D. €4761.47
- E. €4768.47

Q2. You sell 2000 shares on Friday which incurs a €15.95 dealing charge. Your total proceeds amount to €4664.05. By what percentage had the price risen since yesterday's close?

- A. 2%
- B. 3%
- C. 4%
- D. 5%
- E. 6%

Q3. The difference between the buying price and the selling price on a given day is known as 'the spread'. If the percentage spread is calculated as a percentage of the higher value, which day has the largest percentage spread?

- A. Monday
- B. Tuesday
- C. Wednesday
- D. Thursday
- E. Friday

Set 18

The Planets		
Planet	Distance from Sun (million km)	Orbital Period (Earth Days)
Mercury	58	88
Venus	101	224.7
Earth	150	365.25
Mars	228	687

Q1. As a planet travels around the Sun (orbital period), the distance it covers is 6.3 times its distance from the Sun. At what speed does the Earth travel around the Sun in km/sec?

- A. 0.03
- B. 0.3
- C. 3
- D. 30
- E. 300

Q2. How much faster than Venus is Mercury travelling? The distance travelled by a planet around the Sun is the distance of the planet from the Sun multiplied by 6.3.

- A. 24%
- B. 47%
- C. 67%
- D. 85%
- E. 125%

Q3. A future astronaut on Mars has to wait at least 8m 40s to get a response from Earth to his messages. This minimum delay occurs when the distance of Mars from Earth is the distance of Mars from the Sun minus Earth's distance from the Sun. At what speed do his messages travel (in km/sec)?

- A. 120,000
- B. 150,000
- C. 210,000
- D. 270,000
- E. 300,000

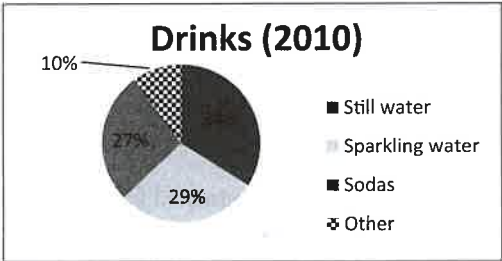
Set 19

Belgium (Population 10.84m)	
	Percentage of Population
Under-18	31.2
Have a driver's licence	64.7
Employed	42.7
Never been outside Belgium	2.9
Watch more than 10h of TV per week	59.8

- Q1. If 98.3% of the people who have never been outside Belgium are under 18 years old, how many people over 18 have never been outside Belgium?
- A. 10
B. 186
C. 798
D. 5344
E. 9827
- Q2. Employed people are 25% less likely to watch 10h or more of TV in a week. What percentage of the non-employed people watch more than 10h per week of TV?
- A. 38.1
B. 59.8
C. 68.8
D. 74.2
E. 84.8
- Q3. At least what percentage of non-employed people have a driver's licence based on the data in the table?
- A. 22%
B. 29.7%
C. 35.1%
D. 38.4%
E. 44.7%

Set 20

Bakery Sales (€'000)		
	2009	2010
Bread	86.3	87.2
Pastries	27.9	30.1
Pies/Pasties	18.6	18.8
Drinks	43.7	44.9



- Q1. In 2008 bread sales were 7% less than in 2010. By how much did they rise in 2009?
- A. 3.9%
B. 4.7%
C. 5.9%
D. 6.4%
E. 6.8%
- Q2. In 2010 sparkling water sales rose 10% from 2009. What was the value of sparkling water sales in 2009?
- A. €11,837
B. €11,881
C. €16,228
D. €18,493
E. €21,276
- Q3. What was the difference between sales of water (sparkling + still) and pastries in 2010?
- A. €782
B. €992
C. €1230
D. €1684
E. €1813

Set 21

Customer Survey 2010		
	Responses	Average (values are rounded to the nearest 1 decimal place)
Greeting	99	8.1
Ambiance	93	8.3
Cleanliness	90	7.7
Quality	102	6.9
Value for money	88	6.8
Overall satisfaction	110	7.2
Note: Averages that are closer to a lower value are rounded down and all other values are rounded up.		

Q1. There were 110 surveys returned but not all questions were answered on every card. What was the average number of responses per question?

- A. 99
- B. 97
- C. 50
- D. 6.1
- E. 5.3

Q2. Questions 1 and 6 both got the same number of total marks when all their scores were added up. How many was this (rounded to the nearest whole number)?

- A. 792
- B. 795
- C. 797
- D. 800
- E. 802

Q3. The average mark for overall satisfaction dropped 10% from its 2009 value. What was the overall satisfaction score in 2009?

- A. 6.5
- B. 7.2
- C. 7.7
- D. 7.9
- E. 8.0

Set 22

Off Licence Sales		
	Percentage of sales by value	
	2009	2010
Beer	29.1	30.4
Wine	28.8	29.2
Spirits	12.1	10.6
Tobacco	11.8	10.3
Miscellaneous	18.2	19.5

Q1. In 2010, sales went up by 5% on 2009. What was the increase in sales of beer between 2009 and 2010?

- A. 1.5%
- B. 5.2%
- C. 6.7%
- D. 8.1%
- E. 9.7%

Q2. Sales of tobacco were €122 thousand in 2009 and increased by 0.3% in 2010. What was the total value of all sales in 2010?

- A. €1.073 million
- B. €1.188 million
- C. €1.201 million
- D. €1.224 million
- E. €1.278 million

Q3. In 2010, by how much would the sales of wine need to increase to equal those of beer in the same year?

- A. 1.2%
- B. 1.9%
- C. 3.9%
- D. 4.1%
- E. 4.4%

Set 23

Vaccinations in Europe				
Country	Population in 2010 ('000)	Percentage of Population Vaccinated		
		MMR	DPT	Polio
Estonia	1339	87.3	83.2	49.6
Italy	60098	94.8	93.7	51.9
Netherlands	16653	96.2	97.3	54.6
UK	61899	91.6	92.2	46.2

Q1. How many more people were vaccinated against MMR in Italy than in the UK?

- A. 274
- B. 2740
- C. 27,400
- D. 274,000
- E. 2,740,000

Q2. In 2005, the population of the Netherlands was 5% lower than in 2010 but the percentage of the population vaccinated against Polio was 2 percentage points higher. How many more or fewer people were vaccinated against Polio in 2005 than in 2010?

- A. 333,000 fewer
- B. 139,000 fewer
- C. No change
- D. 139,000 more
- E. 333,000 more

Q3. What is the minimum possible number of people who have been vaccinated against both MMR and DPT in Estonia in 2010?

- A. 1.3m
- B. 1.17m
- C. 1.11m
- D. 944K
- E. 55K

Set 24

Shoe Shops in Europe (per 100,000 population)				
	Population in 2010 ('000)	2000	2005	2010
France	62139	8.3	8.4	8.6
Italy	60098	9.3	9.6	9.5
Netherlands	16653	8.2	8.4	8.4
Spain	45317	7.3	7.6	7.2

Q1. How much higher than the lowest per capita rate is the highest per capita rate of shoe shops in the countries shown across all three years?

- A. 2.4%
- B. 12.5%
- C. 25%
- D. 33.3%
- E. 50%

Q2. If the population of France increased by 5% each five years, how many more shoe shops were there in France in 2010 than in 2000?

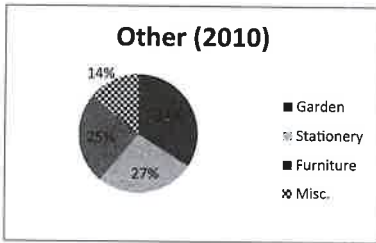
- A. 381
- B. 598
- C. 666
- D. 742
- E. 848

Q3. If the population of Spain increased by 7% between 2000 and 2005, what was the percentage increase in the number of shops over that time?

- A. 4.1%
- B. 7.0%
- C. 11.1%
- D. 11.4%
- E. 12.3%

Set 25

Supermarket Sales (€m)		
	2009	2010
Food	6.62	6.84
Clothing	1.98	2.04
Electrical	1.87	1.99
Other	2.11	2.23



Q1. What was the percentage rise in revenue in the electrical department between 2009 and 2010?

- A. 3.9%
- B. 4.7%
- C. 5.9%
- D. 6.4%
- E. 6.8%

Q2. In 2010, sales in the garden department rose by 4% from 2009. What was the garden department's value of sales in 2009?

- A. €344,000
- B. €542,000
- C. €633,000
- D. €729,000
- E. €802,000

Q3. Stationery sales fell by 3.2% between 2009 and 2010. What percentage of total sales did stationery account for in 2009?

- A. 1.98%
- B. 4.94%
- C. 12.91%
- D. 21.66%
- E. 27.62%

Set 26

Chocolate Consumption in Europe ('000 tonnes)			
	Population 2005 ('000)	2005	2010
Belgium	10415	175	183
Estonia	1346	16.6	17.5
Hungary	9973	131	138
Italy	58645	932	989
Switzerland	7441	129	133

Q1. Which country has the highest per capita consumption of chocolate in 2005?

- A. Belgium
- B. Estonia
- C. Hungary
- D. Italy
- E. Switzerland

Q2. The population of Hungary increases by 8.3% between 2005 and 2010. What is the percentage change in per capita consumption of chocolate over that time?

- A. +5.31%
- B. +1.97%
- C. No change
- D. -2.73%
- E. -5.31%

Q3. In Switzerland, 78% of consumption is of homemade chocolate, which accounts for 12% of total home production, the remainder being exported. How much chocolate did Switzerland export in 2005?

- A. 20,000 tonnes
- B. 198,00 tonnes
- C. 738,000 tonnes
- D. 839,000 tonnes
- E. 928,00 tonnes